Arista Access Node WHI-ACC-61 QoS Congestion Event Major Alarm SOP

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_3394

Description

The QoS congestion event major alarm ALARM_3394 on Arista Access Node WHI-ACC-61 at Whitefield East POP is a critical failure that warrants immediate attention to prevent network instability and potential packet loss. The root cause of this alarm is typically related to Microbursts, interface tail drops, or QoS queue starvation, all of which are indicative of a congested network or misconfigured QoS policies. This SOP outlines the necessary steps to diagnose and resolve this issue on the Arista Access Node, a device that plays a critical role in the network's access layer. The Arista Access Node, specifically the WHI-ACC-61 model, employs a robust hardware platform and operates on top of a feature-rich EOS (Extreme Operating System) that provides an array of QoS features to ensure high-performance, low-latency, and guaranteed packet delivery. However, when faced with congestion events, the QoS policies on the Arista Access Node may not be adequate to handle the increased packet load, leading to tail drops, packet loss, and ultimately, the ALARM_3394 major alarm. To prevent network downtime and data loss, it is essential to promptly diagnose and resolve this issue by identifying the root cause of the congestion event, adjusting the QoS policies, and verifying the resolution.

Pre-requisite Checks

- Confirm that the power supply to the Arista Access Node WHI-ACC-61 is stable and there are no environmental issues at the Whitefield East POP.

- Verify that there are no other alarms or issues on the same node or adjacent nodes that may be contributing to the congestion event.

- Check the redundant link status to ensure that the link is up and functioning correctly.

Potential Root Causes

Cause	Probability
Microbursts causing QoS queue overflows	High
Misconfigured QoS policies leading to queue starvation	Medium
Hardware related issues on the interface causing tail drops	Low

Diagnostic Commands & Expected Outputs

```
Command: show traffic interface Ethernet1/1, Expected Output:
Interface Ethernet1/1, Hardware is Ethernet, Description: The 'show
traffic interface' command is used to analyze the traffic statistics
on the specified interface. It provides details on the packets
transmitted, received, and dropped on the interface, helping to
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identify potential issues with packet loss or congestion.

Command: show qos policy-map interface Ethernet1/1, Expected Output:
Policy-map is attached to the interface Ethernet1/1, Description: The
'show qos policy-map interface' command is used to display the QoS
policy-map configuration on the specified interface. It provides
information on the policy-map, including the packet classification,
marking, and queuing parameters, helping to identify potential issues
with QoS policy misconfiguration.

Command: show interface statistics Ethernet1/1, Expected Output:
Interface Ethernet1/1, 12345 packets received, 12345 packets
transmitted, Description: The 'show interface statistics' command is
used to display the interface statistics, including the number of
packets received and transmitted on the interface. It helps to
identify potential issues with interface congestion or packet loss.
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Resolution Steps

Step 1: Step: Adjust the QoS policy-map configuration to ensure sufficient bandwidth allocation for the affected traffic class, Command: configure terminal policy-map policy-name class class-name bandwidth 100, Description: Adjust the QoS policy-map configuration to ensure sufficient bandwidth allocation for the affected traffic class. This may involve increasing the bandwidth allocation or adjusting the queue parameters to ensure that the traffic is not oversubscribed.

Step 2: Step: Verify that the QoS policy-map configuration is correctly applied to the interface, Command: show qos policy-map interface Ethernet1/1, Description: Verify that the QoS policy-map configuration is correctly applied to the interface to ensure that the traffic is classified and queued correctly.

Step 3: Step: Re-run the diagnostic commands to verify that the issue is resolved, Command: show traffic interface Ethernet1/1 show qos policy-map interface Ethernet1/1 show interface statistics Ethernet1/1, Description: Re-run the diagnostic commands to verify that the issue is resolved and the QoS policy-map configuration is correctly applied to the interface.

Post-Check Verification

```
Command: show traffic interface Ethernet1/1, Expected Output:
Interface Ethernet1/1, 0 packets received, 0 packets transmitted,
Description: The 'show traffic interface' command is used to verify
that the issue is resolved and the traffic is flowing correctly on the
interface.

Command: show qos policy-map interface Ethernet1/1, Expected Output:
Policy-map is correctly attached to the interface Ethernet1/1,
Description: The 'show qos policy-map interface' command is used to
verify that the QoS policy-map configuration is correctly applied to
the interface and the traffic is classified and queued correctly.

Command: show interface statistics Ethernet1/1, Expected Output:
Interface Ethernet1/1, 0 packets received, 0 packets transmitted,
Description: The 'show interface statistics' command is used to verify
that the issue is resolved and the traffic is flowing correctly on the
interface.
```

Roll-back Procedures

- Step: Rollback the QoS policy-map configuration to the previous version, Command: configure terminal rollback 1, Description: Rollback the QoS policy-map configuration to the previous version to restore the original configuration and resolve the issue.

- Step: Verify that the issue is resolved after rolling back the configuration, Command: show traffic interface Ethernet1/1 show qos policy-map interface Ethernet1/1 show interface statistics Ethernet1/1,

Description: Verify that the issue is resolved after rolling back the configuration and the QoS policy-map configuration is correctly applied to the interface.